

Type '0' system: $G(s) = \frac{k}{(1+sT_1)(1+sT_2)}$

Put, $s = j\omega$

$$G(j\omega) = \frac{k}{(1+j\omega T_1)(1+j\omega T_2)}$$

Magnitude $|G(j\omega)|$

Take magnitude of numerator and denominator.

$$|1+j\omega T| = \sqrt{1+(\omega T)^2}$$

$$|G(j\omega)| = \frac{k}{\sqrt{1+(\omega T_1)^2} \sqrt{1+(\omega T_2)^2}}$$

Phase angle $\angle G(j\omega)$

phase of each factor: $\angle 1+j\omega T = \tan^{-1}(\omega T)$

Since they are denominator, the phase is negative:

$$\angle G(j\omega) = -[\tan^{-1}(\omega T_1) + \tan^{-1}(\omega T_2)]$$

$\therefore$ Final polar form.

$$G(j\omega) = \frac{k}{\sqrt{1+(\omega T_1)^2} \sqrt{1+(\omega T_2)^2}} \angle -[\tan^{-1}(\omega T_1) + \tan^{-1}(\omega T_2)]$$

At $\omega = 0$.

$$|G(j0)| = k, \quad \angle G(j0) = -0^\circ$$

pole starts at $(k, 0^\circ)$ on the positive real axis.

at $\omega = \infty$ $|G(j\omega)| = 0, \quad \angle G(j\omega) = -180^\circ$

Curve approaches the origin along the (-)ve real axis.

separation of real and imaginary axis -

$$G(j\omega) = \frac{k}{(1+j\omega T_1)(1+j\omega T_2)}$$

$$\begin{aligned}(1+j\omega T_1)(1+j\omega T_2) &= 1 + j^2\omega^2 T_1 T_2 + j(T_1 + T_2)\omega \\ &= 1 - \omega^2 T_1 T_2 + j(T_1 + T_2)\omega\end{aligned}$$

$$\text{So, } G(j\omega) = \frac{k}{(1 - \omega^2 T_1 T_2) + j\omega(T_1 + T_2)}$$

multiplying the numerator and denominator by the conjugate:

$$G(j\omega) = \frac{k [(1 - \omega^2 T_1 T_2) - j\omega(T_1 + T_2)]}{(1 - \omega^2 T_1 T_2)^2 + \omega^2 (T_1 + T_2)^2}$$

$$\text{So, } \operatorname{Re}\{G(j\omega)\} = \frac{k(1 - \omega^2 T_1 T_2)}{(1 - \omega^2 T_1 T_2)^2 + \omega^2 (T_1 + T_2)^2}$$

$$\operatorname{Im}\{G(j\omega)\} = \frac{-k\omega(T_1 + T_2)}{(1 - \omega^2 T_1 T_2)^2 + \omega^2 (T_1 + T_2)^2}$$

Now, important points for sketching the polar plot
at $\omega = 0$ $G(j0) = k$

real part = k , imaginary part = 0 $\therefore$

so, plot starts at $(k, 0)$.

Imaginary axis crossing occurs when $\operatorname{Re}\{G(j\omega)\} = 0$

$$\therefore k(1 - \omega^2 T_1 T_2) = 0$$

$$\Rightarrow 1 = \omega^2 T_1 T_2$$

$$\Rightarrow \omega = \frac{1}{\sqrt{T_1 T_2}}$$

Substituting $\omega = \frac{1}{\sqrt{T_1 T_2}}$ into $G(j\omega)$

so, the denominator,

$$(1 - \omega^2 T_1 T_2) + j\omega(T_1 + T_2) = 0 + j\omega(T_1 + T_2)$$

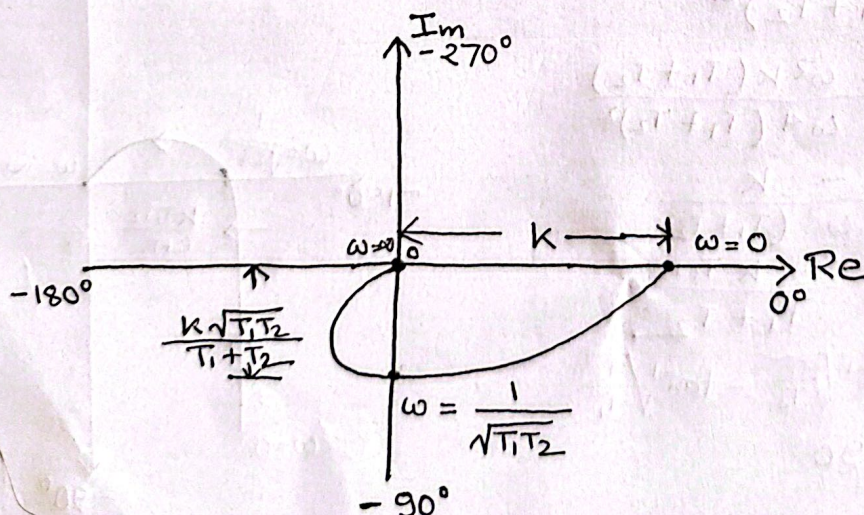
$$\begin{aligned} \therefore G(j\omega) &= \frac{k}{j\omega(T_1 + T_2)} \\ &= \frac{k}{j \times \frac{1}{\sqrt{T_1 T_2}} (T_1 + T_2)} \\ &= -j \frac{k\sqrt{T_1 T_2}}{T_1 + T_2} \end{aligned}$$

$$\begin{aligned} \text{at } \omega = \frac{1}{\sqrt{T_1 T_2}}, \quad \angle G(j\omega) &= - \left[\tan^{-1} \left(\frac{T_1}{\sqrt{T_1 T_2}} \right) + \tan^{-1} \left(\frac{T_2}{\sqrt{T_1 T_2}} \right) \right] \\ &= - \left[\tan^{-1} \frac{\sqrt{T_1}}{\sqrt{T_2}} + \tan^{-1} \frac{\sqrt{T_2}}{\sqrt{T_1}} \right] \end{aligned}$$

$$\text{we know, } \tan^{-1}(a) + \tan^{-1}\left(\frac{1}{a}\right) = 90^\circ = -90^\circ$$

$$\text{at } \omega = \infty \quad \angle G(j\omega) = -180^\circ$$

approaches origin along negative real axis.



Type '1' system: $G(s) = \frac{k}{s(1+sT_1)(1+sT_2)}$

$$G(j\omega) = \frac{k}{j\omega(1+j\omega T_1)(1+j\omega T_2)}$$

$$= \frac{k}{\omega \sqrt{1+\omega^2 T_1^2} \sqrt{1+\omega^2 T_2^2}} \angle -90^\circ - \tan^{-1} \omega T_1 - \tan^{-1} \omega T_2$$

at $\omega=0$; $|G(j\omega)| = \infty$ $\angle G(j\omega) = -90^\circ$

at $\omega=\infty$; $|G(j\omega)| = 0$ $\angle G(j\omega) = -270^\circ$

Separating real and imaginary parts.

$$G(j\omega) = \frac{k}{j\omega(1+j\omega(T_1+T_2)+j^2\omega^2 T_1 T_2)}$$

$$= \frac{k}{j\omega - \omega^2(T_1+T_2) - j\omega^2 T_1 T_2}$$

$$= \frac{k}{-\omega^2(T_1+T_2) + j(\omega - \omega^2 T_1 T_2)} \times \frac{-\omega^2(T_1+T_2) - j(\omega - \omega^2 T_1 T_2)}{-\omega^2(T_1+T_2) - j(\omega - \omega^2 T_1 T_2)}$$

$$= \frac{-\omega^2(T_1+T_2) + j(\omega - \omega^2 T_1 T_2)}{\omega^2(1-\omega^2 T_1 T_2)^2 + \omega^4(T_1+T_2)^2} + \frac{-k\omega(1-\omega^2 T_1 T_2)}{\omega^2(1-\omega^2 T_1 T_2)^2 + \omega^4(T_1+T_2)^2}$$

$\text{Im}\{G(j\omega)\} = 0$

$\Rightarrow -\omega^2 k(T_1+T_2) = 0$

$\Rightarrow -k\omega(1-\omega^2 T_1 T_2) = 0$

$\Rightarrow \omega = \frac{1}{\sqrt{T_1 T_2}}$

Re $\{G(j\omega)\}$'s denominator,

$$\omega^2(1-\omega^2 T_1 T_2)^2 + \omega^4(T_1+T_2)^2$$

$$= 0 + \omega^4(T_1+T_2)^2$$

$$\therefore \text{Re}\{G(j\omega)\} = \frac{-\omega^2 k(T_1+T_2)}{\omega^4(T_1+T_2)^2}$$

$$= \frac{-k}{\omega^2(T_1+T_2)}$$

$$= \frac{-k T_1 T_2}{T_1+T_2}$$

$$\angle G(j\omega) = \angle -90^\circ - \tan^{-1} \sqrt{\frac{T_1}{T_2}} - \tan^{-1} \sqrt{\frac{T_2}{T_1}}$$

$$= \angle -90^\circ - 90^\circ$$

$$= \angle -180^\circ$$

